

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE - 2 (Design)

Architectural Modelling (CO2)

Group No	C1
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Problem Title	Cafe Management System

1. Structural Design

Class Diagram: The following UML class diagram is derived from the current implementation classes: User, AbstractUser, Customer, Admin, Cart, MenuItem, Order, OrderItem, Payment, InventoryItem, and SalesReport. The design reflects the role-based structure introduced in the code, with AbstractUser as the abstract parent class, Customer and Admin as specialised roles, association between User and AbstractUser for role selection, composition between Order and OrderItem, association between Payment and Order, and usage dependencies from Customer/Admin to the supporting classes.

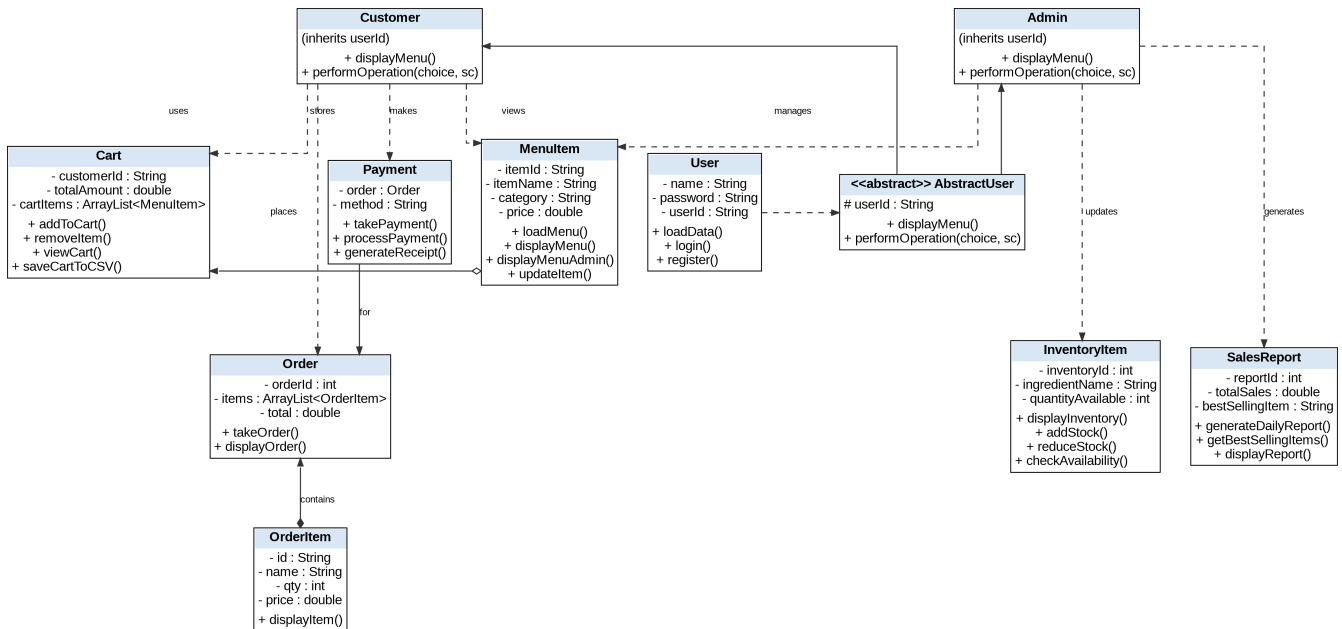


Figure 1. UML class diagram for the Cafe Management System.

2. Behavioural Design

2.1 Actors

Actor	Role in the System
Customer	Registers or logs in, views the digital menu, adds items to the cart, places the order, and makes the payment.
Admin	Logs in, manages menu items, displays inventory, checks stock availability, updates stock levels, and generates the daily report.

2.2 Use Cases

No.	Use Case	Description
1	Register	Customer creates a new account and receives a customer ID.
2	Login	Customer or admin enters valid credentials and accesses the system.
3	Display Menu	Menu items are displayed with category, price, and availability.
4	Add to Cart	Customer selects menu items and stores them in the cart.
5	Place Order	Customer confirms the selected items and prepares order details.
6	Make Payment	Payment is processed and a receipt can be generated.
7	Update Menu Item	Admin edits price or availability, adds a new item, or removes an item.
8	Display Inventory	Admin views the inventory details stored in the inventory file.
9	Check Inventory	Admin checks whether an inventory item is available or at reorder level.
10	Add / Reduce Stock	Admin updates stock quantity by adding or reducing items.
11	Generate Daily Report	Admin generates the daily sales report showing total sales and best-selling item.

2.3 Use Case Diagram

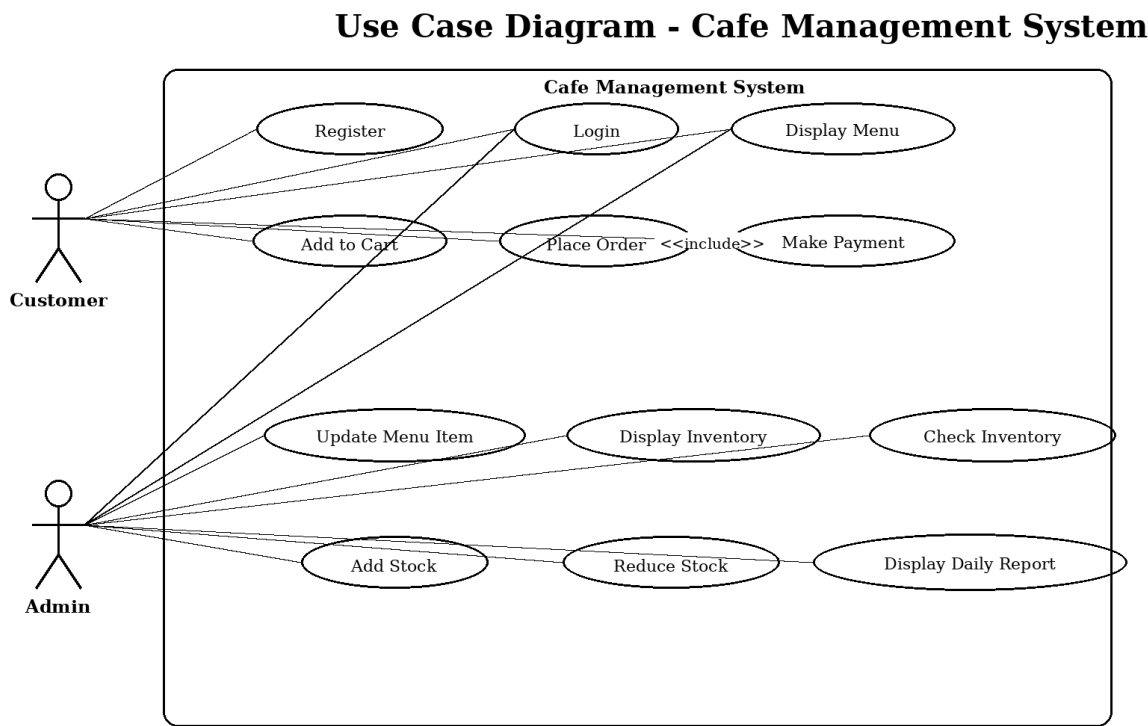


Figure 2. Use case diagram for the Cafe Management System.

3. Sequence Diagrams

3.1 Key Use Case 1 - Customer Login, Cart and Payment

Scenario: The customer registers or logs in, views the menu, adds items to the cart, places the order, makes payment, and receives the receipt.

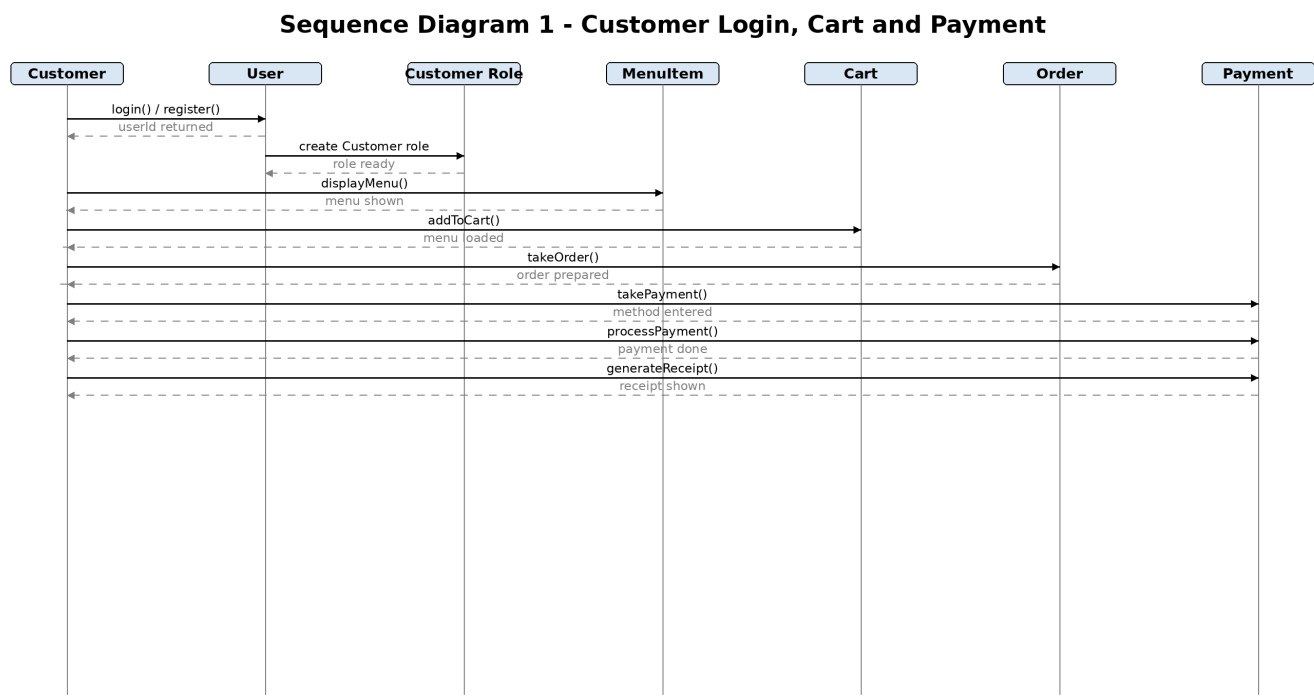


Figure 3. Sequence diagram for customer login, cart and payment.

3.2 Key Use Case 2 - Admin Inventory and Daily Report

Scenario: The admin logs in, displays or updates menu items, checks inventory availability, adds or reduces stock, and finally generates the daily sales report.

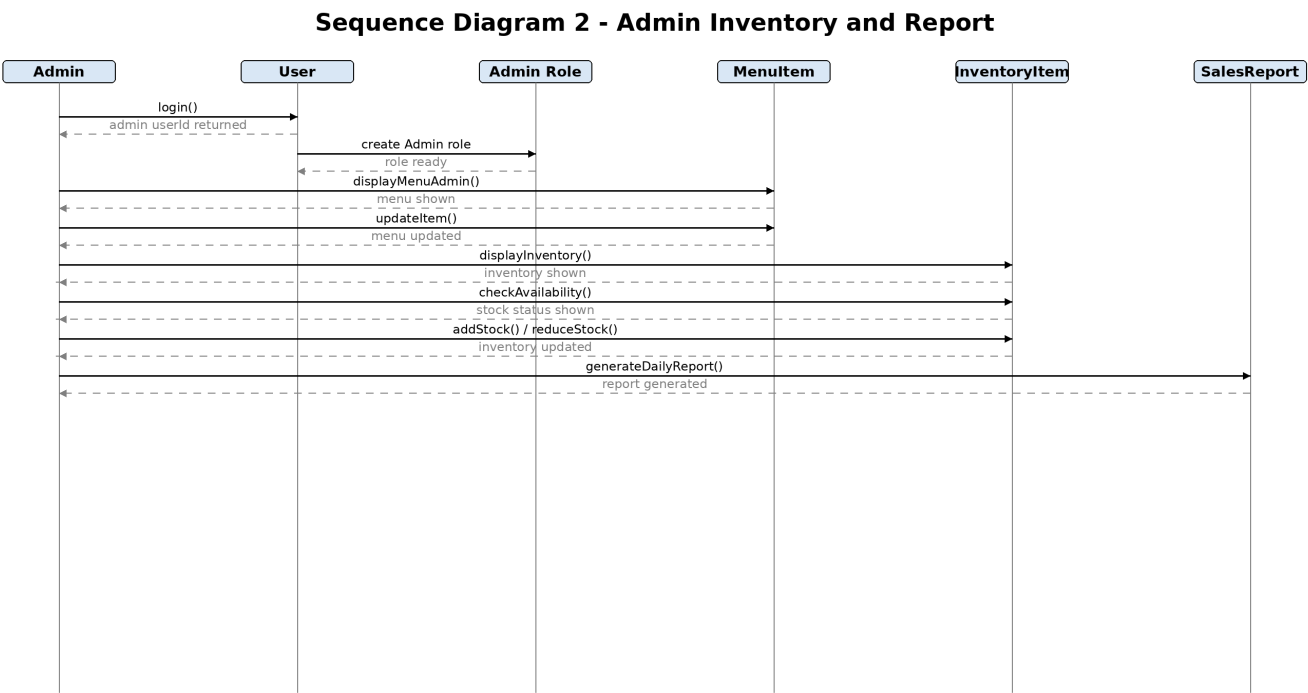


Figure 4. Sequence diagram for admin inventory and daily report.

4. Design Summary

This design models both the structure and behaviour of the Cafe Management System. The class diagram shows the current implementation classes and their relationships, the use case diagram shows the interaction between actors and system functions, and the sequence diagrams explain the message flow for the most important operations.

- The design also shows how authentication, role selection, menu handling, cart operations, order processing, inventory management, and reporting are connected to the main customer and admin interactions in the system.